Reinforcement Learning for Optimal Control

Reinforcement Learning, HW6–HW8 and Comprehensive Exercise Q2

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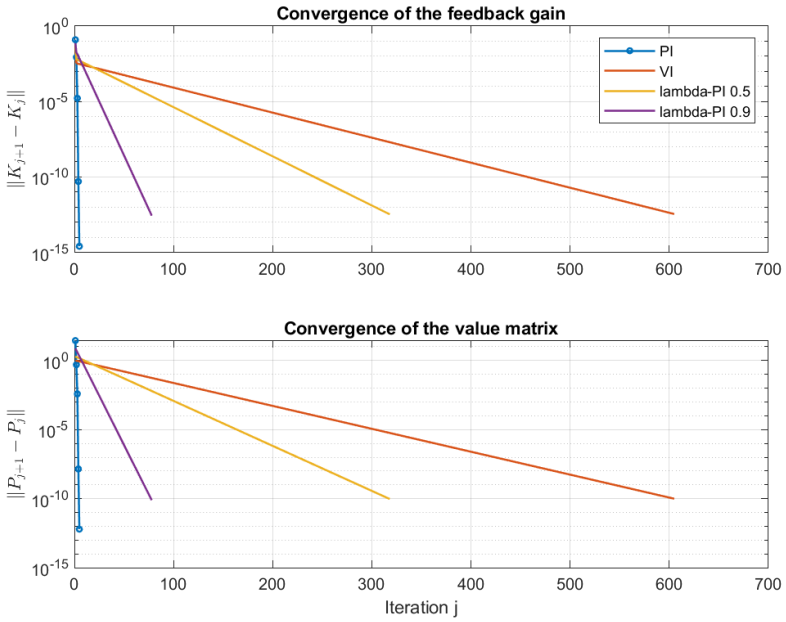
Each problem below has an exact answer, so every learned controller is compared with it.

1. Discrete-time LQR (HW6)

Plant $x_{k+1} = Ax_k + Bu_k$ with three states, $Q = I_3$, $R = 1$, $u = -Kx$. From $K_0 = 0$ (the plant is stable):

- **Policy iteration:** $(A - BK_j)^\top P_{j+1}(A - BK_j) - P_{j+1} + Q + K_j^\top RK_j = 0$, then $K_{j+1} = (R + B^\top P_{j+1}B)^{-1}B^\top P_{j+1}A$.
- **Value iteration:** $P_{j+1} = (A-BK_j)^\top P_j(A-BK_j)+Q+K_j^\top RK_j$ with the same improvement step.
- **λ-policy iteration:** $P_{j+1} = (1-\lambda)A_j^\top P_j A_j + \lambda A_j^\top P_{j+1}A_j + K_j^\top RK_j + Q$, $A_j = A - BK_j$; $\lambda = 1$ is PI and $\lambda = 0$ is VI.
- **Model-free Q-learning:** $Q_K(x,u) = [x;u]^\top H[x;u]$ is fitted from measured transitions with $z_k^\top Hz_k - z_{k+1}^\top Hz_{k+1} = x_k^\top Qx_k + u_k^\top Ru_k$, and $K \leftarrow H_{uu}^{-1}H_{ux}$.

All reach the DARE gain $K^* = [-0.06432,\ -0.06989,\ 0.06668]$: PI in 5 iterations, $\lambda = 0.9$ in 78, $\lambda = 0.5$ in 318, VI in 605, Q-learning in 4 (from data only, error 6×10^{-15}).



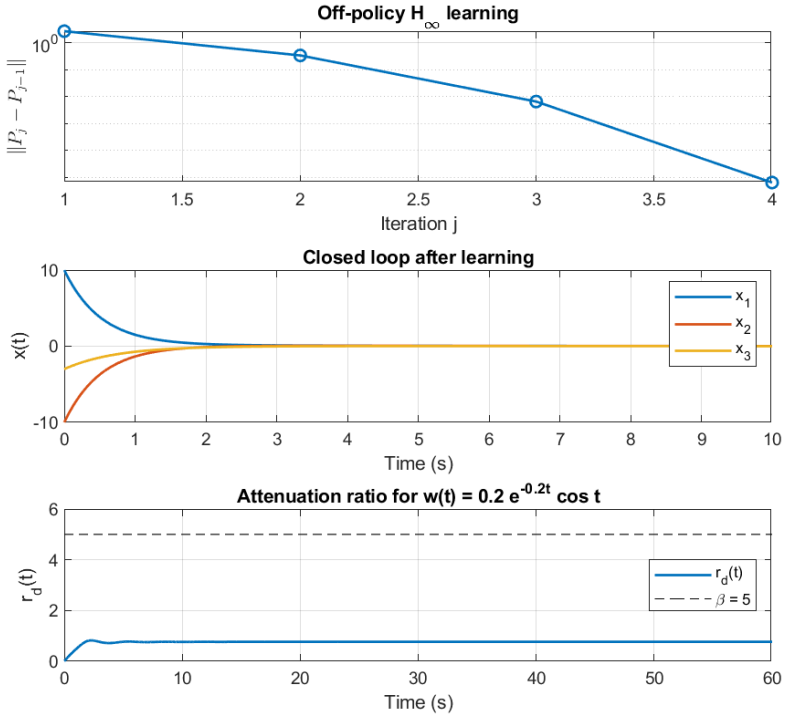
2. Integral RL and off-policy H_∞ control (HW7)

IRL. For $\dot{x} = Ax + Bu$ and $u = -K_jx$, $V_j = x^\top P_j x$ satisfies $x(t)^\top P_j x(t) - x(t+T)^\top P_j x(t+T) = \int_t^{t+T} x^\top(Q + K_j^\top RK_j)x\,d\tau$. P_j is fitted from measured states (no A), then $K_{j+1} = R^{-1}B^\top P_j$. It reached the CARE gain $[-0.13523,\ -0.15010,\ 0.43293]$ in 4 iterations.

H_∞ **control** with disturbance d and $\beta = 5$: the reference is the game Riccati equation $A^\top P + PA + Q - PB_1 R^{-1} B_1^\top P + \beta^{-2} PB_2 B_2^\top P = 0$. With data from an exploring behaviour policy and target policies $u_j = -K_{1j}x$, $d_j = K_{2j}x$,

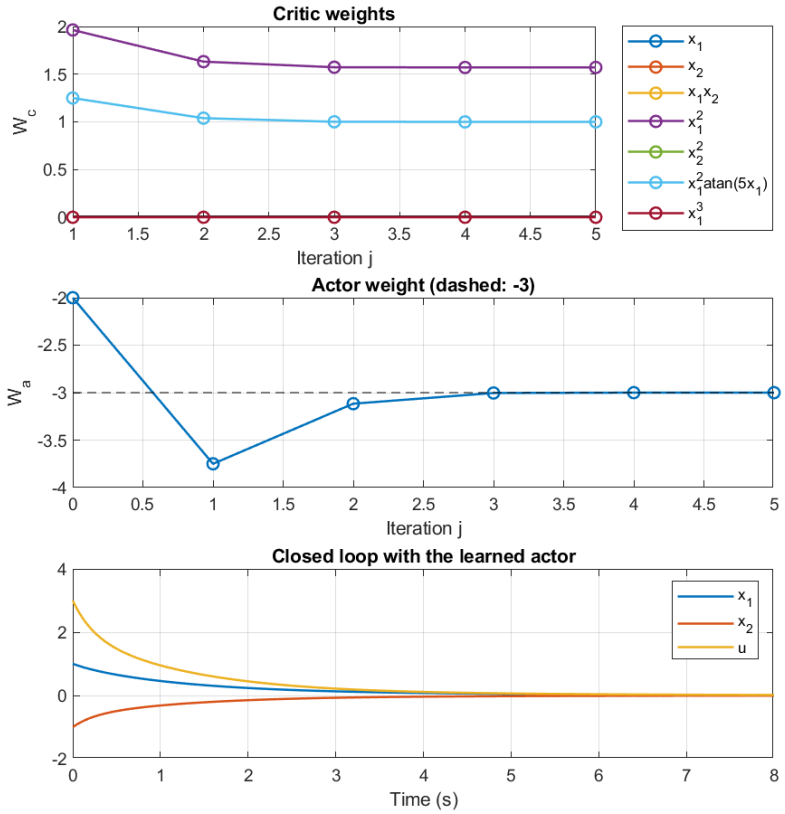
$$x^\top P_j x\Big|_t^{t+T} = \int_t^{t+T} [-x^\top (Q + K_{1j}^\top R K_{1j} - \beta^2 K_{2j}^\top K_{2j})x + 2(u + K_{1j}x)^\top R K_{1,j+1} x + 2\beta^2 (d - K_{2j}x)^\top K_{2,j+1} x]\, d\tau,$$

which is linear in $(P_j, K_{1,j+1}, K_{2,j+1})$ and needs no model. It matched the game Riccati solution in 3 iterations ($\|P - P^*\| = 2 \times 10^{-5}$). For $w(t) = 0.2e^{-0.2t}\cos t$ the measured attenuation ratio is 0.77, below $\beta = 5$.



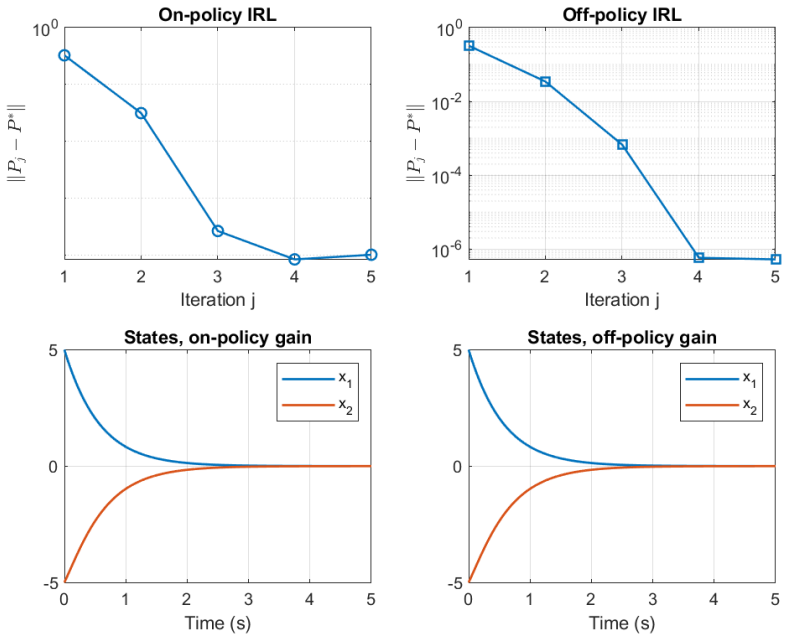
3. Actor-critic for a nonlinear system (HW8)

For $\dot{x}_1 = x_2$, $\dot{x}_2 = -x_1(\frac{\pi}{2} + \arctan 5x_1) - \frac{5x_1^2}{2(1+25x_1^2)} + 4x_2 + 3u$ with $Q = \mathrm{diag}(0,1)$, $R = 1$, the optimum is $V^* = x_1^2(\frac{\pi}{2} + \arctan 5x_1) + x_2^2$, $u^* = -3x_2$. With critic basis $\phi = [x_1,\ x_2,\ x_1x_2,\ x_1^2,\ x_2^2,\ x_1^2 \arctan 5x_1,\ x_1^3]$ and actor $u = W_a x_2$, the continuous-time actor-critic alternates a critic fit of $W_c^\top(\phi(x(t)) - \phi(x(t+T))) = \int r\, d\tau$ and an actor fit to $-\frac{1}{2}R^{-1}g^\top \nabla\phi^\top W_c$. Starting from $W_a = -2$ it reached $W_c = [0, 0, 0, \pi/2, 1, 1, 0]$ to four decimals and $W_a = -3.00003$ in 4 iterations.



4. On-/off-policy IRL and tracking (Comprehensive Q2)

For the unstable plant $A = \begin{bmatrix} 0.5 & 1.5 \\ 2 & -2 \end{bmatrix}$, $B = [1; 4]$, both IRL variants reach the CARE gain [1.6180, 0.9001]. Off-policy IRL (one batch of behaviour data, no model) reaches $\|P - P^*\| = 6 \times 10^{-7}$; on-policy IRL stalls near 10^{-4} because its probing noise enters the Bellman equation as a bias, which the off-policy form avoids. For tracking $x_d = [\sin t; \cos t]$ I used the augmented state $X = [e; x_d]$ and the discounted Bellman equation $X(t)^\top P X(t) = \int_t^{t+T} e^{-\gamma(\tau - t)} r \, d\tau + e^{-\gamma T} X(t{+}T)^\top P X(t{+}T)$. The learned gains match the discounted ARE to about 10^{-3} for $\gamma = 0.4$ and 0.9; the tracking error stays near 0.7 because the discounted problem with $R = 3$ does not force it to zero.


On-policy IRL
Off-policy IRL
$\|P_j - P^*\|$
Iteration j
States, on-policy gain
States, off-policy gain
x1
x2
Time (s)